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# Morphological determination of pore connectivity as a function of pore size using serial sections

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## Summary

The geometry of pore space in soil is considered to be the key in understanding transport of water, gas and solute. However, a quantitative and explicit characterization, by means of a physical interpretation, is difficult because of the geometric complexity of soil structure.

Pores larger than 40 μm within two soil horizons have been analysed morphologically on 3-dimensional digital representations of the pore space obtained by serial sections through impregnated specimens. The Euler-Poincaré characteristic has been determined as an index of connectivity in three dimensions. The pore connectivity is quantified as a function of the minimum pore diameter considered leading to a connectivity function of the pore space. Different pore size classes were distinguished using 3-dimensional erosion and dilation. The connectivity function turned out to differentiate between two soil materials. The pore space in an upper Ah horizon is intensely connected through pores between 40 and 100 μm, in contrast to the pore space in the AhBv beneath it. The morphological pore-size distributions were compared to the pore-size distribution obtained by water retention measurements. The discrepancy between these different methods corresponds to the expectation due to pore connectivity.

## Introduction

The transfer of water, gas and solutes through soil is governed mainly by the soil's structure. It is an old challenge to soil scientists to relate direct features of soil structure to these processes, so as to predict the behaviour of a given material just from its geometric reality without expensive and complex experiments. Following such a morphological path, the characterization of the geometrical properties of the structure should be quantitative to allow the results to be incorporated into models of the processes. As another requirement the interpretation of the results towards the physics of the processes should be possible. The latter was critical in previous attempts of relating structure and function (Bouma *et al.*, 1977; Walker & Trudgill, 1983; Vogel & Babel, 1994).

Regarding transfer processes, soil structure can be simplified to a two-phase system consisting of solid material and void. One important geometrical aspect is the distribution of the pore volume over a range of effective pore diameters which corresponds to the hydraulic diameter according to the capillary rise equation. This is difficult to accomplish, based on 2-dimensional images of the structure typically used for morphometrical analysis. There are different size criteria in 2D-image processing (Moran, 1994) but none of them with a clear physical meaning. Also stereological methods require strong assumptions about the shape of the pores (Vogel & Babel, 1994). These difficulties can be overcome when the 3-dimensional reality of the structure becomes available by techniques such as serial sectioning and tomography.

Connectivity is another geometrical feature of pore space which is essential in terms of transport in porous media but difficult to measure because, again, the knowledge of the 3D-reality is indispensable. DeHoff *et al.* (1972) presented a method for counting the basic topological quantities using serial sections, namely the volumetric number of disconnected parts of a structural unit, $N_V$, and the volumetric genus, $G_V$, the number of its redundant connections. Macdonald *et al.* (1986*a,b*) applied these concepts to determine the genus of the pore space in Berea sandstone, and they studied the effect of sample size using artificial structures. Principally, an exact determination of the topological quantities requires a global approach, i.e. the complete structure should be known. This is currently not feasible for natural porous media. Consequently, estimates obtained from finite samples lead to maximal values if the pores are assumed to be connected outside the sample. Conversely, the assumption that the pores are not connected leads to a minimal value. Macdonald *et al.* (1986*b*) demonstrated this effect and concluded that increasing the number

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of serial sections which leads to a more isometric volume decreases the edge effect. Scott *et al.* (1988*a,b*) calculated $N_V$ and $G_V$ of crack patterns in soil by tracing the pores along serial sections. They found that 10 to 20 sections sufficed to get stable results.

Recently, Gundersen *et al.* (1993) showed how to get an unbiased estimation of the 3D-Euler-Poincaré characteristic from a pair of parallel sections, a minimum 3-dimensional sample called the disector. The Euler-Poincaré characteristic is defined as $N_V - G_V$ and therefore can be considered to be a measure of connectivity that becomes increasing by negative with increasing connectivity and increasing by positive with decreasing connectivity. This approach is a local analysis where edge effects can be accounted for. However, no estimates of the absolute values of $N_V$ and $G_V$ are obtained. Vogel & Kretzschmar (1996) translated the counting rule of Gundersen *et al.* (1993) into an algorithm for digital image processing and presented first results for two soil horizons considering pores larger than 50 μm. The present paper is an expansion to this work.

A connectivity function is introduced which is defined as the Euler-Poincaré characteristic dependent on the smallest pore diameter considered for the analysis. To get a measure of pore size, the basic operators of mathematical morphology, erosion and dilation, where applied to digital 3D representations of the pore space obtained by serial sections. The methods are illustrated for an artificial structure, a cubic packing of spheres, and applied to the two soil horizons already investigated by Vogel & Kretzschmar (1996). Additionally, the pore-size distributions of the different soil materials are derived from their water retention characteristic and the results are compared to the pore-size distribution obtained by morphological analysis.

The procedures presented require 3-dimensional sets of data on soil structure, which are still hard to get – either by serial sectioning which is tedious or by tomography which is expensive. However, the results have the potential to give an input to modelling processes in soil.

## Methods

### *Material and sample preparation*

Two horizons (Ah and AhBv) of a Terra fusca (Eutric Cambisol) in southern Germany under deciduous forest (beech) were investigated. It is the same soil as studied previously by Vogel & Babel (1994), Vogel (1994) and Vogel & Kretzschmar (1996) who describe the two horizons in more detail. Both horizons are well aggregated, with a clay content of 43% in the Ah (0–10 cm) and 52% in the AhBv horizon (10–40 cm). Due to faunal activity, the aggregates are smaller and more loosely packed in the upper Ah (1.05 g $cm^{-3}$ bulk density) than in the lower AhBv (1.27 g $cm^{-3}$). Binary images of pore space in both horizons are shown in Figure 1.

Undisturbed samples were taken using Kubiena boxes (four replicates, 40 mm × 60 mm × 80 mm) for morphological investigations and cores samples (six replicates, 5 cm in diameter, 4 cm length) for the measurement of the water retention characteristics.

Water in the samples was exchanged by acetone, and then the samples were impregnated with a polyester resin (Vestopal 120L). The dehydration largely preserves the field structure of the soil (Moran *et al.*, 1989). A small block (30 mm × 30 mm × 5 mm) was cut vertically from each impregnated specimen. The surface of the block was ground using a high precision device and then polished using a diamond paste. Thereafter, the surface was etched with 10% hydrofluoric acid for 4 minutes to increase the contrast between solid and void (Vogel & Kretzschmar, 1996). Subsequently the surface was photographed with a digital camera (Kodak DCS200) resolving 1523 × 1011 pixel with the scale set to 13 μm per pixel. Thus, the size of the resulting images was about 20 mm × 13 mm. The photographs were stored as 8-bit grey level images.

Serial images were prepared by eroding the surface by 40 μm using the grinder, followed by the procedure just described: polishing, etching and digitization. In this way, 20 serial surfaces separated by a distance of 40 μm were obtained from each specimen. The first four surfaces of two series from the different soil horizons are shown in Figure 1.

It is essential for further analysis that the relative position of adjacent surfaces is known. This was approximately achieved by mounting the camera on a support tightly connected to the sample holder. There, the polished block could be installed exactly in the same position each time a surface was photographed.

### *Digital image processing*

The grey level images were transformed to binary representations of solid and void using the bilevel segmentation algorithm presented by Vogel & Kretzschmar (1996). Two thresholds are calculated enclosing a narrow fuzzy region on the grey scale where the clear association of the corresponding pixel to either solid or void is not possible due to the grey level only. Within this region the discrimination of pore and solid is based on the neighbourhood relationship of each pixel. The thresholds were calculated for each individual image according to its grey level histogram. This method proved to be especially suited for the detection of pore edges, which is important for tracing the connectivity of pore space along serial images.

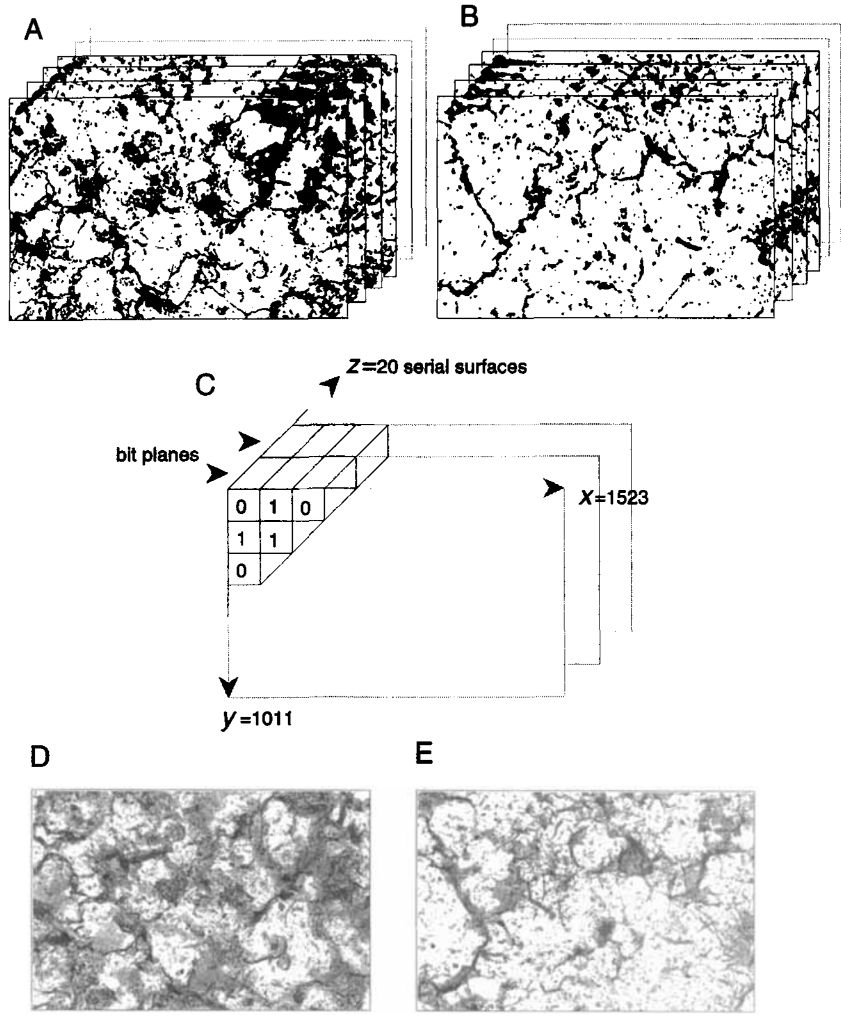


**Figure 1** 20 serial binary images of pore space in the Ah (A) and the AhBv horizon (B) with a resolution of 1523 × 1011 pixel are combined in a computer to a 3D matrix of voxels represented by single bits (C). Images D and E are projections of the serial images where the grey level is calculated as $g(x, y) = \frac{255}{20} \sum_{i=1}^{20} z_i(x, y)$.

The precision of the relative position of serial surfaces could be controlled through the projection of adjacent binary images. Thereby even a slight displacement of an image in relation to its neighbour was detectable. This was the case for about 10% of the images. Evidently, the photographic set-up described above is sensitive when working with a resolution of 13 μm. To avoid the loss of complete series of images as a result of this inaccuracy an attempt was made to correct the displacement. The squared deviations between adjacent images were calculated as

$$D(a, b) = \sum_{x=1}^{N_x} \sum_{y=1}^{N_y} [f(x, y) - g(x + a, y + b)]^2, \qquad (1)$$

where $(x, y)$ are the pixel coordinates, and $f(x, y)$ and $g(x, y)$ are the binary functions of two adjacent images with a total number of $N_x \times N_y$ pixel each. To get the vector $\vartheta(a, b)$ for minimum deviation $D(a, b)$ was minimized, where $a$ and $b$ are integers from the interval [15, −15]. This correction of the shift is based on the assumption that the structure is isotropic in the horizontal plane. The validity of this assumption concerning the present material could be concluded from the small absolute values of $a$ and $b$, which were found to be between −1 and 1 for 90% of the parallel images, while the corresponding vectors showed no preferred direction.

For efficient calculation of pore size distribution and connectivity, each set of 20 binary serial images was stored after shift-correction in a single file representing a 3-dimensional matrix of voxels. Each pixel within the plane of the serial surfaces is coded by three bytes (24 bits) and the subsequent serial images were stored in the subsequent bit-planes. Thus, each voxel is represented by a single bit, being 1 for pore and 0 for solid (Fig. 1).

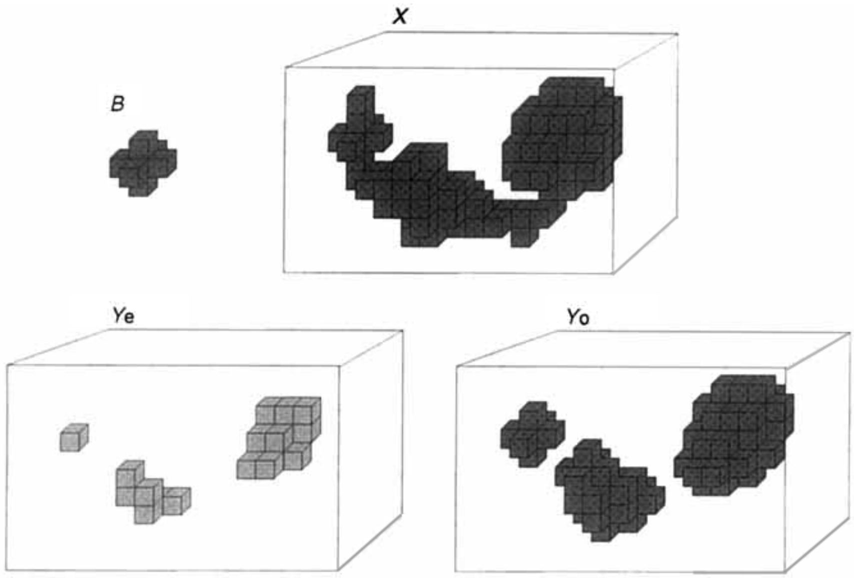


**Figure 2** 3-dimensional opening of a set $X$ by a structuring element $B_x$ centered at each point $x$ within the investigated volume. The eroded set $Y_e = \{x : B_x \subset X\} = X \ominus B$ is subsequently dilated to get the opened set $Y_o = \{x : B_x \cap Y_e \neq \emptyset\} = Y_e \oplus B$ where $\ominus$ and $\oplus$ are the operators for erosion and dilation respectively.

*Pore size distribution*

What is the (morphological) size of a pore? There are many possible answers to this question. Most of them are related to 2-dimensional sections: e.g. the area size distribution of intersected pores (Murphy *et al.*, 1977), the chord size distribution (Jongerius *et al.*, 1972), the distribution of a so called equivalent diameter (diameter of the circle with surface area equal to the surface of an intersected pore) (Bouma *et al.*, 1977) or the starlength (McBratney & Moran, 1990). These measures are difficult to interpret in terms of the 3-dimensional reality and thus do not admit of any physical interpretation. Stereological methods account for 3-dimensional interpretation of 2D-measures (Weibel, 1979). For describing pore space in soil, however, they require strong assumptions on the 3D-shape of the objects (Vogel *et al.*, 1993).

Disregarding the capabilities of existing image analyzers one could define the size of a pore at any point within the pore space as the diameter of the largest ball which includes this point and which fits completely into the pore space. This appears as an intuitive definition of pore size which also reflects the principle of the water retention characteristics of a porous medium. The opening-size distribution defined in mathematical morphology is based on a comparable definition. Mathematical morphology is essentially a set-theoretical approach to image processing. It was formally introduced by Serra (1982). The two basic operations illustrated in Fig. 2 are erosion and dilation of a given set, e.g. for the set of pore voxels within a given volume. Erosion followed by dilation of the eroded set is called opening. Thereby, the original set is intersected with a structuring element of defined size and shape. As shown in Fig. 2, opening removes those parts of the structure which are 'smaller' than the structuring element. Opening with spherical structuring elements of different diameters leads to the opening-size distribution corresponding to the intuitive definition of pore size introduced above. These operations are standard in image processing. They are applied most frequently to 2-dimensional images (Gonzales & Woods, 1992), although there is no additional complexity associated with the 3-dimensional case (Meyer, 1992).

The 3D representation of the pore space obtained from the combination of serial surfaces provides a suitable data base for the determination of the opening-size distribution. In the 3D data set pore space is represented on a rectangular anisotropic grid with a resolution of 13 μm in the $x$-$y$-direction and 40 μm on the $z$-axis normal to the serial surfaces. The spherical structuring elements used for the opening procedure must be defined on a grid having the same geometry and, hence, only a rough approximation to the spherical shape can be realized. In Fig. 3 structuring elements with different radii are shown. They are calculated according to the Euclidian distance of grid elements to the center of the structuring element. Because of the anisotropic grid, the 'sphere' with radius 1 to 3 occupies only one $z$-plane, and there is a jump to the adjacent $z$-planes when increasing the radius from 3 to 4 units. As will be demonstrated below, the opening-size distribution is quite sensitive to such a jump since erosion by a sphere with radius 4 has a relatively strong effect in comparison to the erosion by a sphere with radius 3.

To diminish this effect the resolution of the grid was doubled in $z$-direction by interpolating an additional image between adjacent serial surfaces. The interpolation was done as follows: The binary images of two adjacent surfaces representing the sets of pore voxels $X_1$ and $X_2$ respectively were united in one 2-dimensional image to get the sets $X_1 \cap X_2$ and $X_1 \cup X_2$. For each point $x \in [X_1 \cup X_2]$ four directions (0°, 45°, 90°, 135° to the horizontal axis) were considered and each point $x$ was attributed to the interpolated set $Y_i$ which was closer to the set $X_1 \cap X_2$ than to the solid phase in any of the considered directions. The solid

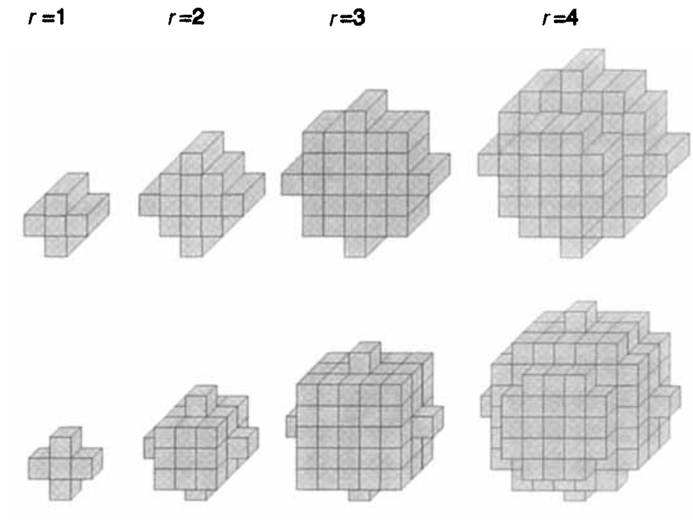


**Figure 3** Structuring elements with different radius $r$ [number of voxels in x-y-direction] approximated to a sphere on a 3-dimensional rectangular grid with different resolution in z-direction: $x \times y \times z = 13 \times 13 \times 40$ units (upper row) $13 \times 13 \times 20$ units (lower row).

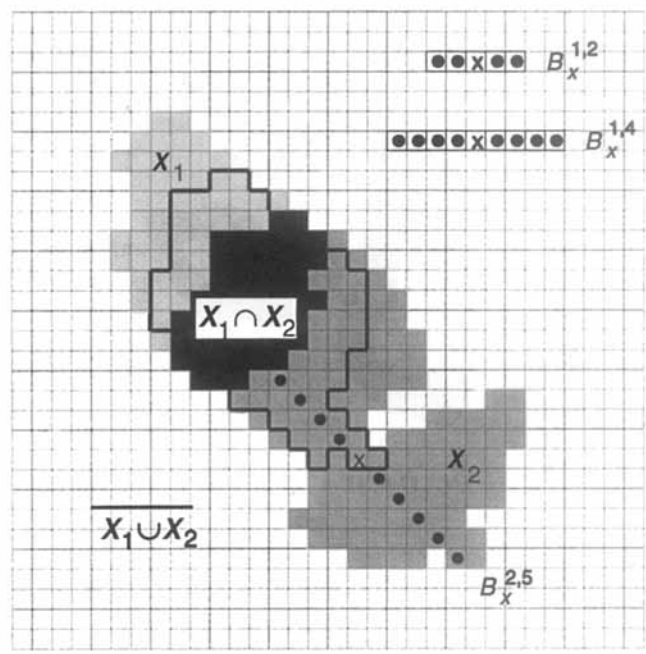


**Figure 4** Interpolation of an intermediate image between two parallel surfaces containing the sets $X_1$ and $X_2$ using a 1-dimensional structuring element $B_x^{n,m}$ centered at each point $x \in X_1 \cup X_2$ ($n = 1, 2, 3, 4$ indicating the angle of rotation – 0°, 45°, 90°, 135° – to the horizontal axis and m [number of pixels] the length). In accordance with Equation (2), all pixels are attributed to the interpolated set (enclosed by the thick line) which were closer to the set $X_1 \cap X_2$ than to the background ($\overline{X_1 \cup X_2}$) in at least one of the directions considered.

phase is represented by the complement of the pore space $\overline{X_1 \cup X_2}$. This procedure, illustrated in Fig. 4, does not change the topological properties of the structure. It can be formalized by introducing a 1-dimensional structuring element $B_x^{n,m}$ centred at each point $x \in X_1 \cup X_2$ where $n = 1, 2, 3, 4$ indicates the rotation and $m \in Z$ the length of $B$. Then, the interpolated set $Y_i$ is defined as

$$Y_i = \{x : \exists[B_x^{n,m} \cap [X_1 \cap X_2] \neq \emptyset \wedge B_x^{n,m} \cap [\overline{X_1 \cup X_2}] = \emptyset]\} \; \forall x \in \{X_1 \cup X_2\}, \tag{2}$$

where $\wedge$ is the logical AND.

The 3-dimensional opening was performed on the 3D voxel matrix including the interpolated surfaces. The spherical structuring elements with different radii were calculated for the changed grid geometry (Fig. 3). The radii $r$ of the structuring elements are expressed in pixel units of 13 μm corresponding to the resolution in the $x$-$y$ plane of the grid. Openings were calculated for $r = 1, 2 \ldots, 7$ leading to structuring elements with diameters $d = (2r + 1)13$ μm between 40 μm and 195 μm. The volume density of pore space was obtained after each opening from the proportion of pore voxels. Thereby, the volume of pores larger than the diameter of the structuring element was determined.

In addition to the morphological approach, the pore size distribution was obtained from the water retention characteristics measured on the undisturbed soil cores. After saturation, the samples were equilibrated on a ceramic plate at successive potentials of 1, 3, 5 and 10 kPa corresponding to equivalent pore diameters of 300, 100, 60, and 30 μm. The water content was measured

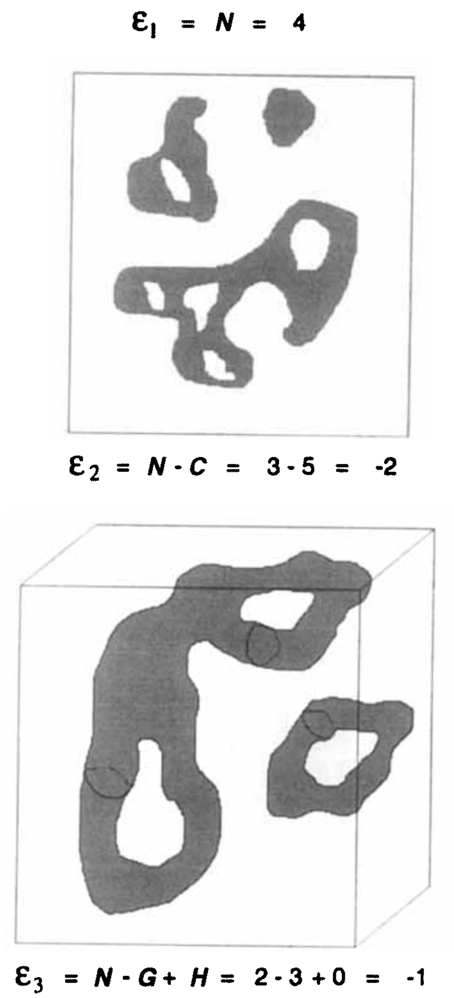


**Figure 5** Euler-Poincaré characteristic $\varepsilon_n$ in one, two and three dimensions. $N$ = number of disconnected components, $C$ = number of redundant connections, $G$ = genus (corresponding to C in 2D) and $H$ = completely enclosed cavities (hard to draw).

gravimetrically at each potential. The total pore volume was calculated from bulk density and particle density. The latter was measured with the pycnometer method (Blake & Hartge, 1986).

*Pore connectivity*

The connectivity of a geometric body is characterized by its geometrical topology which can be described in different dimensions by the Euler-Poincaré characteristic $\varepsilon_n$ (Hadwiger, 1957) (Fig. 5). Considering a closed set $X$ in 1-dimensional space, $\varepsilon_1(X)$ is the number of segments of $X$ on a straight line. In 2D, $\varepsilon_2(X)$ is the number $N$ of separated components of $X$ minus the number $C$ of redundant connections within $X$ where the latter equals the number of holes in $X$:

$$\varepsilon_2(X) = N(X) - C(X). \tag{3}$$

In 3D the number of redundant connections is called the genus $G$. $\varepsilon_3(X)$ is defined as

$$\varepsilon_3(X) = N(X) - G(X) + H(X), \tag{4}$$

where $H(X)$ is the number of completely enclosed cavities in $X$.

The Euler–Poincaré characteristic can be interpreted as a measure of connectivity. Increasing positive values indicate decreasing connectivity of the structure, and decreasing negative values indicate increasing connectivity. An unbiased estimate of $\varepsilon_3$ can be obtained with the disector (Gundersen *et al.*, 1993) which is a pair of closely spaced parallel 2-dimensional sections. Vogel & Kretzschmar (1996) applied this method using digital image processing. In contrast to the direct determination of $N$, $G$ and $H$,

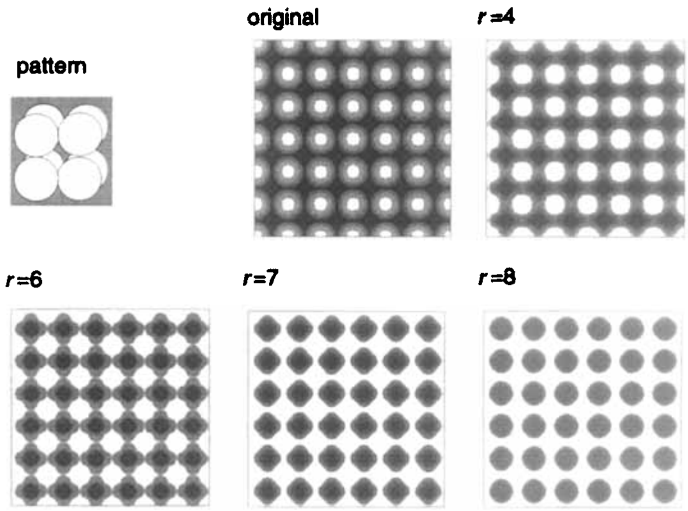


**Figure 6** Projections of a cubic packing of spheres realized on a rectangular grid ($x \times y \times z = 13 \times 13 \times 40$ units, pores are dark): Original and resulting structures after opening with spherical structuring elements of different radius $r$ [number of voxels in $x$-$y$ direction]. Note that the pore space at $r = 6$ is still connected through one pixel.

where edge effects are of great importance (DeHoff *et al.*, 1972), the estimation of $\varepsilon_3$ with the disector is a local measurement. Here, the problems of edge effects are completely resolved by considering the additivity properties of $\varepsilon$ (Prasad *et al.*, 1989).

The four sets of 20 serial surfaces from each soil horizon were considered as $n = 19$ adjacent disectors $i$ to calculate $\varepsilon_{3,i}$ of the pore space according to Vogel & Kretzschmar (1996):

$$\varepsilon_3 = \tfrac{1}{2}[\varepsilon_2(X_1) + \varepsilon_2(X_2) - 2\varepsilon_2(X_1 \cap X_2)], \tag{5}$$

where $X_1$ and $X_2$ are the sets of pore pixel within the two parallel images of a disector and $X_1 \cap X_2$ is obtained through the logical AND operator applied to these images. The formulation in Equation (5) follows directly from the formulation given by Vogel & Kretzschmar (1996). Equation (5) demands that the distance between the two parallel images of the disector is smaller than the pertinent elements of the structure.

The volumetric Euler-Poincaré characteristic $\varepsilon_V$ was calculated as

$$\varepsilon_V = \frac{\Sigma_{i=1}^{n} \varepsilon_{3,i}}{V}, \tag{6}$$

where $V = 20\ \text{mm} \times 13\ \text{mm} \times 0.7\ \text{mm} \approx 197\ \text{mm}^3$, is the volume sampled by the serial surfaces. To obtain an estimate of the connectivity as a function of pore size, $\varepsilon_V$ was calculated for the pore space after openings with structuring elements of increasing diameter (see above). In this way the connectivity of the pore space was evaluated several times with the lower limit of the pore size successively increasing by steps of 26 μm from 40 to 195 μm. Hence, the lower limit of the smallest pore size class (40–66 μm) corresponds to the separation of the serial sections. This may lead to underestimation of the connectivity of these pores because connectivity may be lost in some cases when tracing the pores between sections. DeHoff *et al.* (1972) give a guide to the trade-off between size of features and spacing of sections. They recommend that spacing between adjacent sections be smaller than 1/3 of the mean intercept length of the phase of interest. However, the size criteria applied here correspond to the minimum intercept length and, therefore, the risk of loosing connectivity is required only for the smallest class of pores.

In the following the relation between $\varepsilon_V$ and the lower limit of pore size considered will be referred to as the connectivity function.

## Results and discussion

### *Cubic packing of spheres*

To demonstrate the determination of the opening size distribution and the Euler-Poincaré characteristic a well known artificial structure was investigated. A cubic packing of spheres was sampled on a 3-dimensional matrix of rectangular voxels, the geometry of which corresponds to that obtained by the serial surfaces of the soil investigated: $x \times y \times z = 13 \times 13 \times 40$ units. The diameter $\hat{d}$ of the spheres was set to 27 voxels in the $x$-$y$ plane. Taking the unit of measure as μm, leads to $\hat{d} = 351$ μm. The geometry of this simple structure is illustrated in Fig. 6. In the $x$-$y$ plane $6 \times 6$ entire spheres were generated plus half spheres at the edges; in the $z$-direction only one entire sphere was present also connected to half spheres at the edges.

This structure was subjected to morphological openings of the pore space between the spheres using spherical structuring

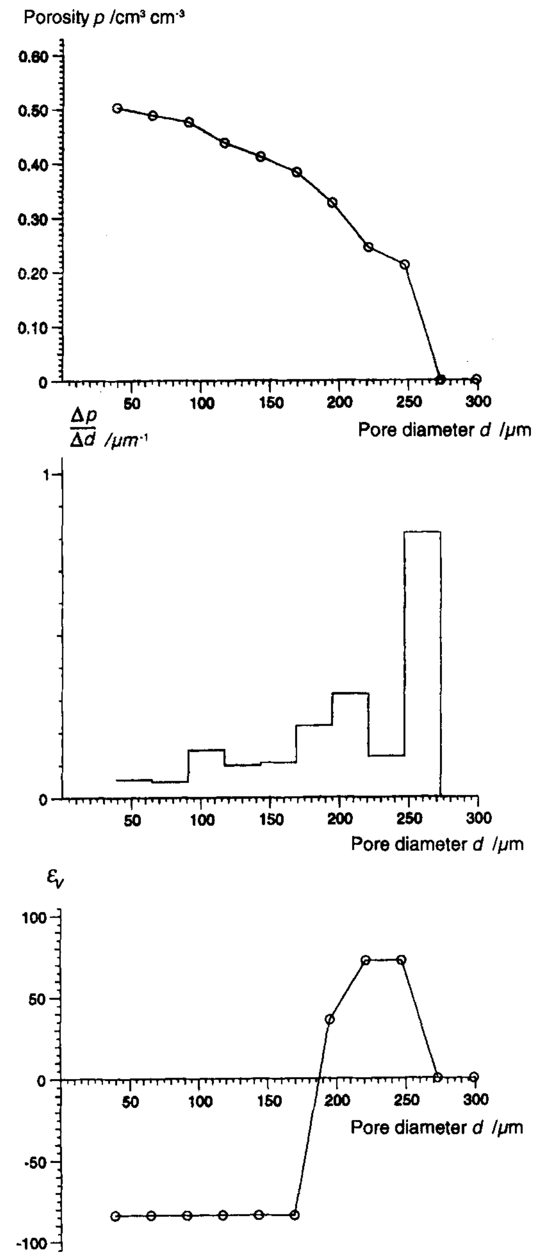


**Figure 7** Porosity $p$ of the cubic packing of spheres versus the minimum pore diameter $d$ considered for the measurement, derived pore-size distribution and the connectivity function. The minimum pore diameter corresponds to the diameter of the structuring element used for opening.

elements of different diameters (3, 5, 7, . . ., 23 voxels). Projected images of the pore space after different openings are shown in Fig. 6. The results of pore volume density versus the opening size and the derived pore size distribution are shown in Fig. 7 together with $\varepsilon_3$ for the pore space.

The curves in Fig. 7 are less smooth than would be expected from a cubic packing of ideal spheres. Clearly, this is due to the anisotropic geometry of the rectangular grid. Regarding the pore-size distribution in Fig. 7 (middle), which is the first derivative of the porosity function (Fig. 7 top), the local maximum at the size class around 100 µm reflects the jump of the thickness of the structuring element when going from a radius of 3 to 4 voxels (see Fig. 3). Also the local minimum at the size class around 230 µm is an artefact of the shape of the structuring elements in relation to the shape of pore space. In this example these effects are intensified because of the uniform shape of the pores. The maximum pore diameter corresponds to the theoretical value of $d_{max}$ = 257 µm which can easily be calculated for the simple geometry of a cubic packing of spheres, $d_{max} = \hat{d}(\sqrt{3}-1)$.

The connectivity of the pore space – expressed by $\varepsilon_3$ – remains constant at a value of −84 for the first six steps of morphological openings. This corresponds to the theoretical value of the modelled pore structure ($\varepsilon_3$ = 1 connected component − 85 redundant connections) where the edges of the investigated volume are considered to be solid. Then, the critical value is

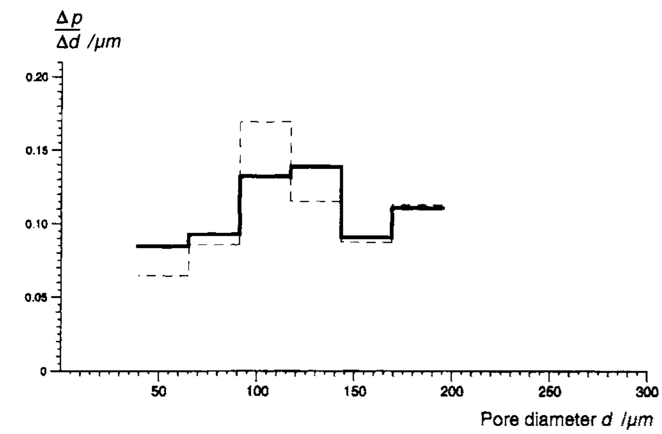


**Figure 8** Opening-size distribution of pores in one sample of the Ah horizon based on the existing 20 serial surfaces only (dashed line) and after interpolating one image between adjacent serial surfaces (solid line).

reached where the connectivity of the pore space is lost. However, $\varepsilon_3$ does not immediately reach the theoretical value of 72 (number of isolated cavities between the spheres). This is again an artefact of the grid geometry. In the *x-y* plane the connectivity is lost first while the pores are still connected in *z*-direction (Fig. 6, $r = 7$), leading to 36 disconnected components.

In a physical context, the critical pore size where the connectivity is lost could be interpreted as the air entry value for the desorption characteristic of the structure. However, in the present example, the air entry value is slightly over-estimated. The theoretical value would be $\hat{d}(\sqrt{2}-1) = 145$ µm whilst the pore space is still connected for pores larger than 169 µm. This is because the opening procedure is based on a local analysis of the structure. Indeed, there is no continuity for a ball with a diameter larger than 145 µm moving through the pore space, but there is a voxel within the neck of the pore space that can be reached with a ball smaller than 169 µm (see Fig. 6, $r = 6$). Consequently, the pore size distribution obtained by morphological opening does not exactly correspond to the hydraulic diameters of a porous medium, but it is a good approximation.

*Soil horizons*

The same procedure as for the cubic packing of spheres was applied to the sets of 20 serial surfaces from the two soil horizons. Here, one image between adjacent surfaces was interpolated to diminish the effect of the anisotropic grid on the determination of pore size. In Fig. 8 the pore-size distributions for one sample from the Ah horizon are shown which were calculated with and without interpolation. As expected, this interpolation has no consequences for pores considerably larger than the separation of the serial surfaces (> *ca* 150 µm). However, the marked maximum obtained for the pores of about 100 µm which was already discussed to be an artefact due to the discrete shape of the corresponding structuring element is significantly reduced through interpolation of one intermediate image.

In Figs 9 and 10 the results of pore-size distribution and connectivity are shown for each of the four samples from the two soil horizons. Figure 11 shows the projected 3D structure of the pore space within the two horizons after different steps of opening. The results of both measurements are drawn into a common graph presuming the equivalence between the morphological pore size obtained by the opening procedure and the hydraulic pore-size obtained by the water retention measurements. For water retention data, the *y*-axis is to be interpreted as air filled porosity, while the *x*-axis denotes the matric potential. On the other hand, the morphological pore volume density is drawn versus the minimum pore diameter considered.

However, an equivalence of morphological and physical results could be expected only if the pore space attributed to a certain size forms a continuum without any necks of smaller diameter. Certainly, this cannot be assumed since the water retention characteristic is hysteretic.

In the Ah horizon the morphological and physical results are clearly different. Although the resolution of the physical data is coarse, it can be concluded that the pore-size classes between 100 and 300 µm were less well represented by the water retention results. Such a discrepancy should be expected for discontinuous pore space where the desaturation of the pores is determined by the diameter of narrow necks.

Moreover, there is some evidence that the air filled porosity at $\Psi = -1$ kPa which was calculated from particle density, bulk density and the water content at this point has been over-estimated. This can be explained by the fact that the volume of entrapped air after saturation of the samples is attributed to this largest pore-size class – whatever the size of the air filled pores.

In the AhBv horizon the pore space is fairly evenly distributed over the different pore-size classes. Compared to the Ah, the volume of pores between 50 and 150 µm is reduced by *ca* 50%. In principle, the pore-size distributions obtained for the AhBv horizon indicate the same discrepancy between the two independent methods although less pronounced than in the Ah horizon.

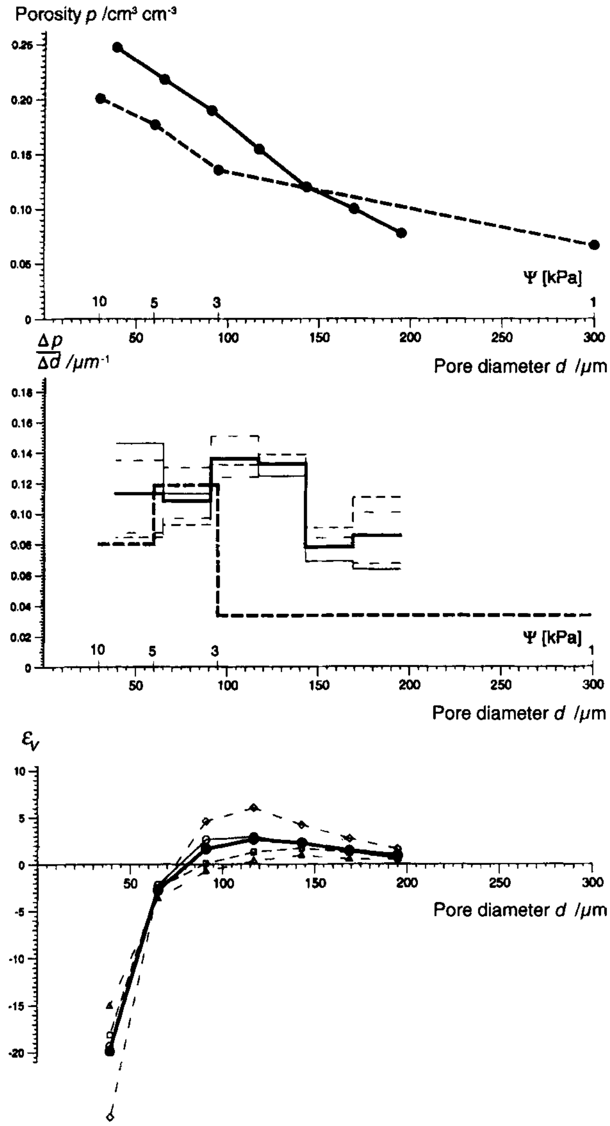


**Figure 9** Top: Porosity $p$ of the Ah horizon versus the minimum pore diameter $d$ considered for the measurement (solid line; average of four replicates) compared to the air filled porosity as a function of the water potential obtained by the water retention measurement (dashed line). Middle: Derived pore-size distribution with the results for all 4 replicates (thin dashed lines). Bottom: Connectivity function of the 4 replicates (dashed lines) and average values (solid thick line).

The connectivity function of the two horizons is shown in Figs 9 and 10. As above the value of $\varepsilon_V$ specifies the number of disconnected parts minus the number of redundant connections per unit volume. Hence, the overall connectivity increases as $\varepsilon_V$ decreases.

To get a better idea of the meaning of the connectivity function let us imagine what should be expected for the curve of this function in soil. From the definition of the connectivity function it should be continuous over the whole size spectrum of pores. Presuming the existence of a maximum pore diameter, the connectivity function should tend to zero at this point. Moreover, presuming that the total pore space in soil is a continuum, its Euler-Poincaré characteristic should be negative. Consequently, the curve of the connectivity function in soil is expected to start at a negative value and to end at zero. However, what happens between these extremes?

Figure 12 shows some examples. The connectivity function may be a straight line if the geometry of the pore space is similar for the existing size range of pores. As another example, there may be a threshold where the connectivity function jumps from negative to positive values, indicating that large pores are connected only by smaller necks. This is the case for the cubic packing of spheres discussed above and may be similar in aggregated soil material. The form of these theoretical connectivity functions is similar to comparable functions which were measured in randomly filled three-dimensional networks by Jernot & Jouannot (1993) and in a two-dimensional fuzzy random model of soil pore structure by Moran & McBratney (1996). In this study only a small section of the connectivity function of the two soil horizons could be investigated (40–200 μm) which, however, is of major importance for fast transport in soil.

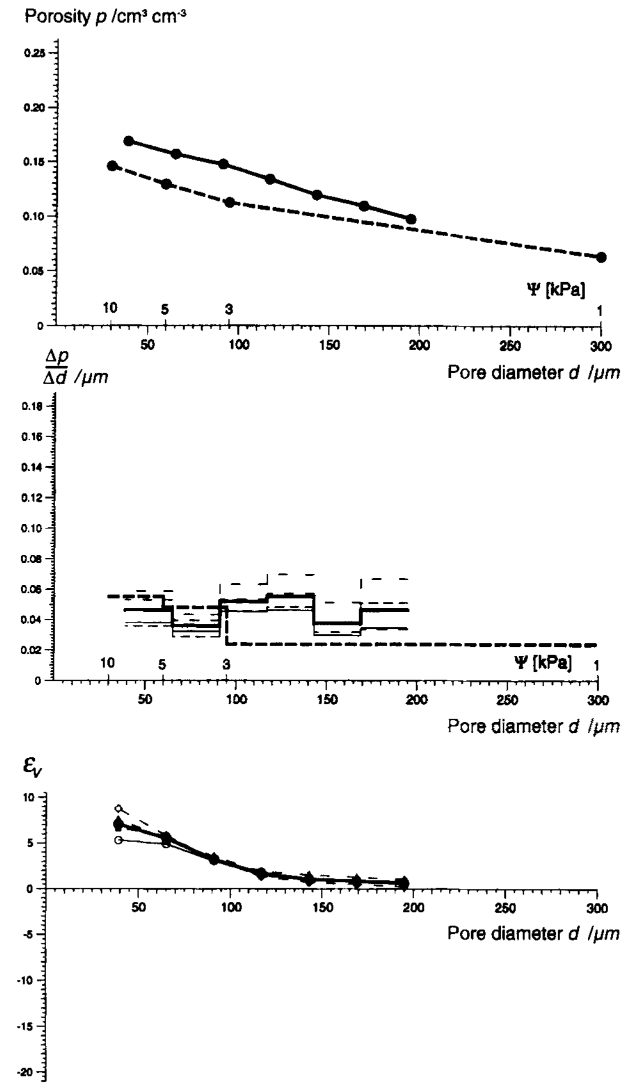


**Figure 10** Top: Porosity $p$ of the AhBv horizon versus the minimum pore diameter $d$ considered for the measurement (solid line; average of four replicates) compared to the air filled porosity as a function of the water potential obtained by the water retention measurement (dashed line). Middle: Derived pore-size distribution with the results for all four replicates (thin dashed lines). Bottom: Connectivity function of the four replicates (dashed lines) and average values (solid thick line).

In the Ah horizon, the large fraction of pores smaller than *ca* 100 μm causes a significant increase in connectivity, which may be considered as a threshold of the connectivity function. This threshold corresponds to the shape of the water retention data in relation to the morphological pore-size distribution indicating necks within the pore space with a diameter of about 100 μm. A quantitative approach to this relation is currently not possible, since the connectivity function does not provide any direct information on the actual volume of the connected or disconnected parts of the pore space. This would require a discrete model of the pore network. In contrast to the upper Ah horizon, the connectivity function of the lower AhBv horizon (Fig. 10) does not show an increasing connectivity with decreasing minimum pore diameter within the size range of pores considered. The two soil horizons are clearly different in their connectivity function.

The variation of the results from the four replicates within each horizon is small. This is true for the pore-size distribution as well as for the connectivity function. The volume of 197 mm$^3$ sampled by the series of 20 serial surfaces appears to be sufficient for the characterization of pores $<$ 200 μm.

The values of $\varepsilon_V$ obtained for the two horizons confirm first results of Vogel & Kretzschmar (1996). There, the disector method was applied to serial sections of the same material using another photographic device. Considering pores larger than 50 μm, the mean values of −10.7 for the Ah and 4.7 for the AhBv horizons were obtained.

The large connectivity in the Ah horizon can be explained from qualitative investigations of the structure (Fig. 11). Large pores, which result probably from earth worm activity, are almost entirely filled with small aggregates of about 60 μm in diameter. These aggregates, mainly cast material of enchytraeids, are loosely packed with a dense network of fine pores in between.

The methods applied here are, in principle, independent of scale, and therefore they can be also applied to data from other

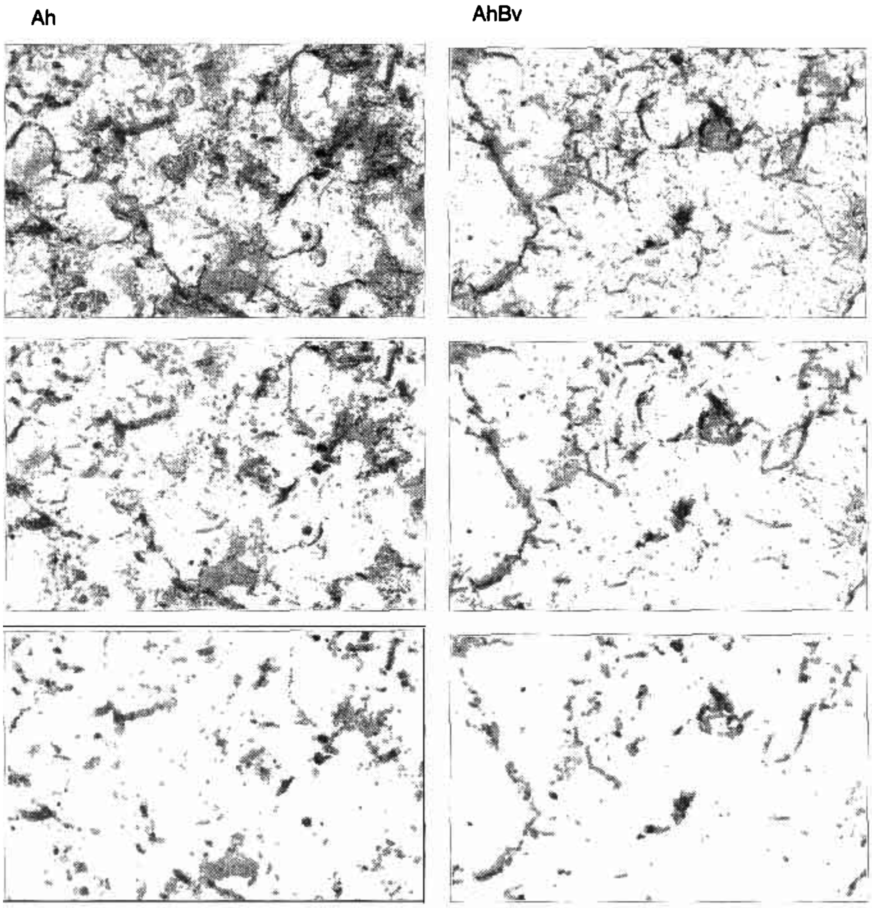


**Figure 11** Projected images of the 3D structure of the two horizons after opening with structuring elements of 65 μm (top), 117 μm (middle) and 169 μm (bottom) in diameter. The images are calculated the same way as described in Figure 1 where the original image is shown.

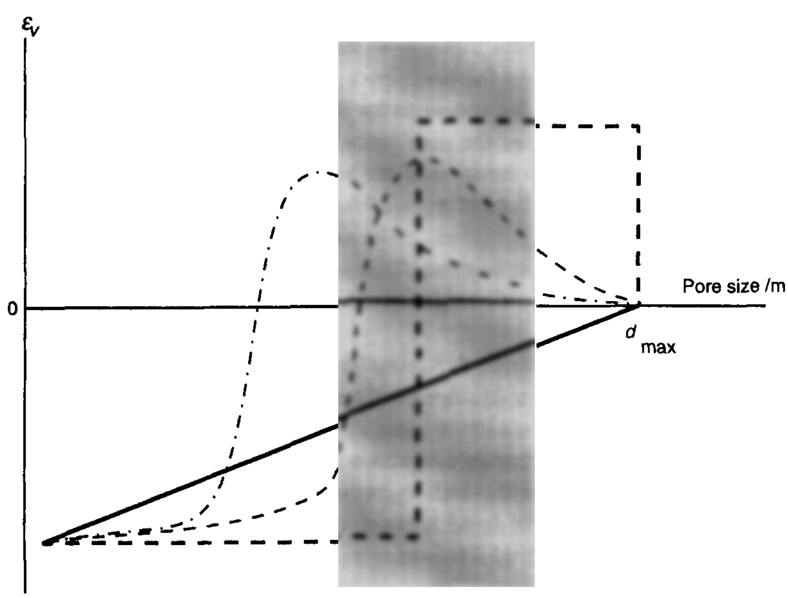


**Figure 12** Four examples of the connectivity function of pore space: a cubic packing of spheres (thick dashed line), pore space with similar geometry of the whole size range of pores (thick solid line) and two hypothetical curves for aggregated soil material (thin dashed lines). In the present study, only a small section of this function could be investigated (grey shaded area).

sources, as for example computer tomography. The link between structural and functional properties of soil remains a challenge. The results presented here are just a small step in this direction.

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